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Step Changes in West of Shetland Offshore Completion Design - First Two Online Multistage Fracs in Clair Deliver Operational Efficiency and Production Uplift

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Abstract

In the Greater Clair Area, the largest oil field in Europe, Hydraulic Fracturing (HF) has demonstrated its value in terms of delivering initial production uplift increases of 3 to 6 times with the first HF wells installed in Clair Phase 1, 2019, and Clair Ridge, 2022. Clair Ridge platform started production in 2018; however, some wells have not encountered sufficient natural fractures (NF) to replicate oil production rates from the initial appraisal wells, increasing the appetite for HF.

SPE 216632 (Bird, 2023) documented successful production delivery from intervention-based HF in Clair, while highlighting these operations were inefficient on platforms not specifically designed to support HF operations. SPE 216632 expressed the case to move away from delivering HF in Clair as intense and complex well work campaigns to instead efficiently deliver HF "online" via drill pipe as part of the lower completion installation (additional operational details captured in SPE-226610 (van Loon, 2025)).

This paper will document the successful planning, design, and execution of the first two "online" HF completion operations delivered for Clair Ridge. 5 HF were executed in well B22 and 6 HF were executed in well B24, including record breaking efficiency for Clair with 4 treatments in 4 days. The process to optimise the completion design will be detailed through completion design and optimization, as well as B22 production outcomes and lessons learned implemented for B24.

The paper will document the potential to incorporate sleeves designed for hydraulic fracturing in addition to multiple additional sleeves designed to provide proppant retention, sand control, and remote activated sleeves for intervention-less production start-up in differing reservoir environments (especially regarding rock strength). Providing such functionality currently increased the number of sleeves per zone (e.g., increasing from 8 to 32 sleeves in an 8 zone well). Developing a sleeve with all functionalities combined into a single sleeve was also assessed. The pros and cons of each approach will be presented outlining the risks associated, such as accidental sleeve shifting or proppant blockage around sand control sleeve annulus.

B22 targeted Clair Unit III which has historically been completed with passive sand control standalone screens (SAS) which was a borderline decision. Ultimately, B22 was the first Unit III well to be completed without SAS, with HF providing additional mitigation for this decision. The process to install a robust sand-

face completion will be documented. Details will be provided of solids production monitoring on startup through a temporary well clean-up package designed to remove solids and water and ensure smooth plant operations, including the use of Cutting Re-Injection (CRI) well to dispose of flow back fluids on the first well, with further optimisations made for the future.

"Online" HF operations have been operationally and technically successful, thus predicted B22 production performance post-drill and actual production post HF will be reviewed, including installing a remote activated downhole choke inserted to control production from a NF zone at risk of water production. Analysis of B22 tracers to compare production splits on well start-up will also be presented.

Introduction

The Clair Oilfield and its reservoir characteristics have been the subject of extensive research and documentation across numerous publications (Ridd, 1981) (Allen, 1992), (Barr, 2007), (Coney, 1993), (Nichols, 2005), (Webster, 2018), (Roy, 2022), (Dumitrache, 2022), (Bird, 2023). For context, a concise overview is provided here. The field was first identified in 1977 through exploration well 206/08-1A, which encountered a substantial oil column within a thick sequence of Devonian to Carboniferous continental sandstones. Spanning over 200 square kilometers and containing an estimated 7 billion barrels of stock tank oil initially in place (STOIIP), Clair stands as the largest oilfield on the UK Continental Shelf. Despite its scale, the field presents considerable reservoir complexity, largely due to intricate natural fracture systems.

A lengthy appraisal phase including a 1996 extended well test demonstrated sustainable production rates of 15,000 to 20,000 barrels of oil per day from long horizontal wells. These wells targeted highly productive, naturally fractured zones aligned with mapped fault structures, proving instrumental in enabling the field's development. Development approval was received in 2001, with production commencing in 2005 via the Clair Phase 1 platform primarily targeting Horst and Graben regions. Clair Ridge platform (online 2018), focuses on the Ridge area (Blue in Figure 1). Collectively, these installations form the Greater Clair Area, which includes both developed and undeveloped reservoir zones across multiple horizons.

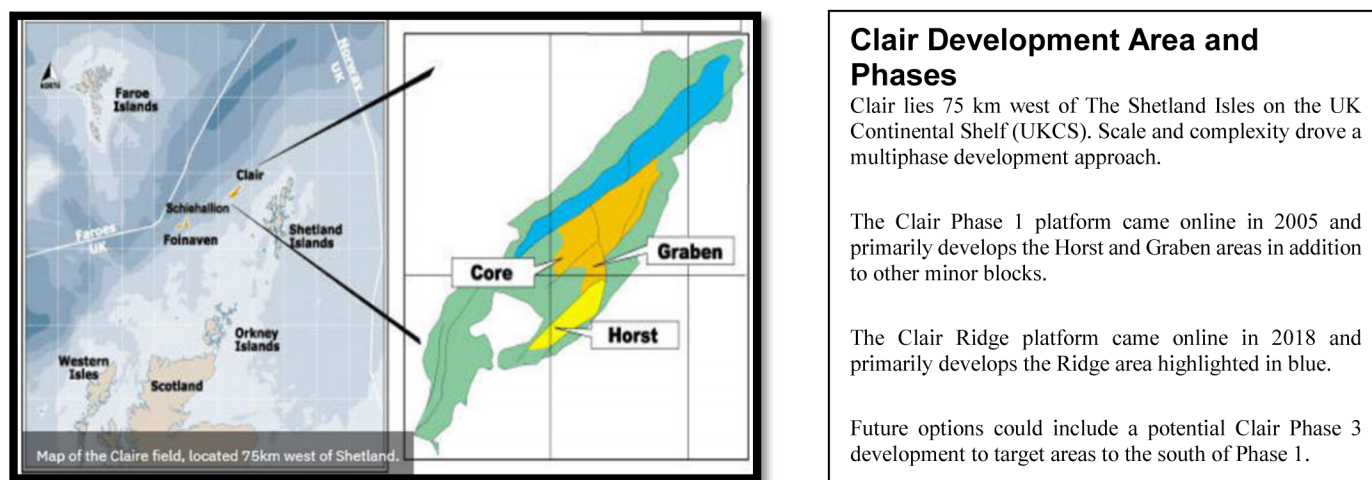


Figure 1—Clair location and primary development areas.

Reservoir Description

The Devonian age Old Red Sandstone (ORS) reservoir lies 1.5-2.5 km below sea level, sealed by Late Cretaceous mudstones and sourced from the Kimmeridge Clay Formation. The two ORS lithostratigraphic units, the Upper (UCG) and Lower Clair Group (LCG). The LCG, containing the bulk of the oil-in-place, is a sandstone characterized by large variations in facies and permeabilities ranging from <1md to several 100md. The LCG is subdivided into six units, I-VI, based on variations in sedimentary facies and

heavy mineral assemblages (Morton, 2003). Development drilling preferentially targets the highest quality reservoirs in Units V and III.

The LCG is defined as a dual-permeability system, with a variable fracture distribution, generally classified as a Type II/Type III fracture reservoir (Nelson, 1981) (Figure 2). Reservoir deliverability and connectivity are primarily controlled by the distribution of existing natural fracture networks. Three main scales of fracture features are observed in core: fault related, fracture corridors unrelated to large offsets and lower density joint-type networks. Open fracture density varies considerably with highest densities in Units V and VI. Open fracture orientation varies between structural compartments (Barr et al., 2007). However, laterals in both Units V and VI have penetrated extensive sections of unfractured or low-fracture-density rock.

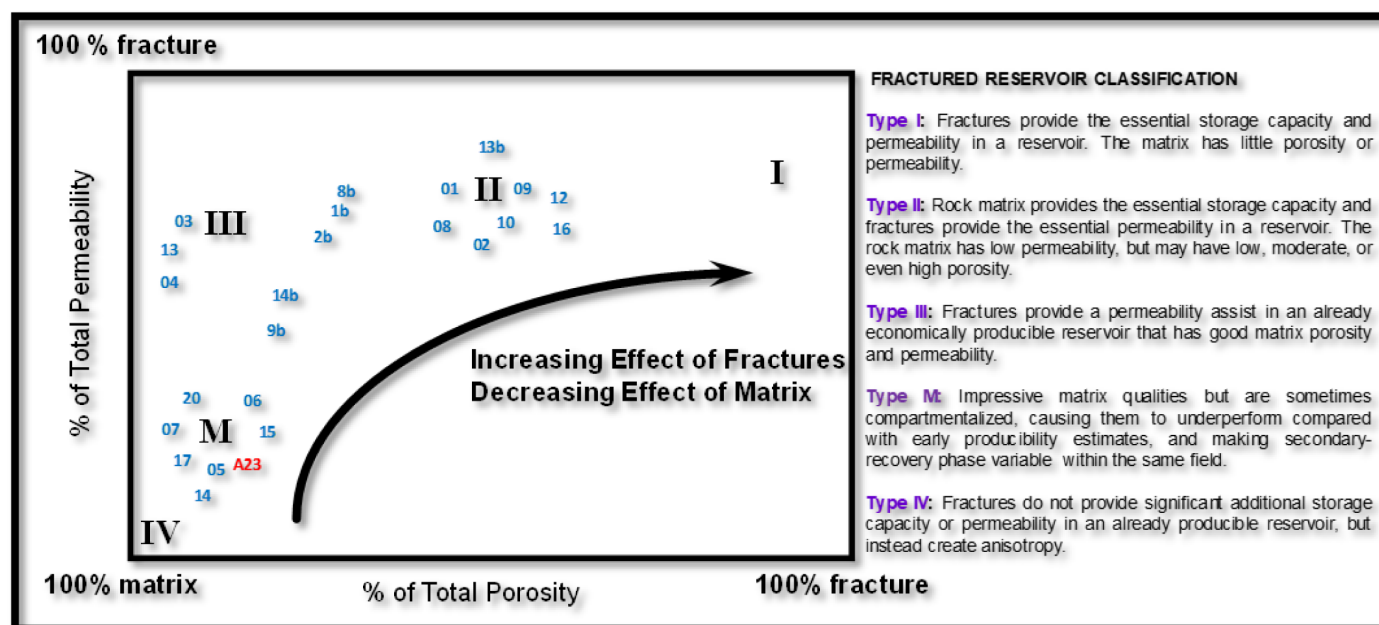


Figure 2—The Clair Field wells categorized as per the fracture classification (in blue wells number).

Further details can be found in SPE 250250 (Roy, 2022).

Field development phases and hydraulic fracturing progress

The scale and complexity of the Clair reservoir drove a phased multiplatform development (Figure 1 and Figure 3). Newer wells incorporate historic lessons ensuring robustness to dynamic uncertainties in this naturally fractured reservoir (NFR). Development wells have encountered varying degrees of natural fractures, the primary control on production outcomes.

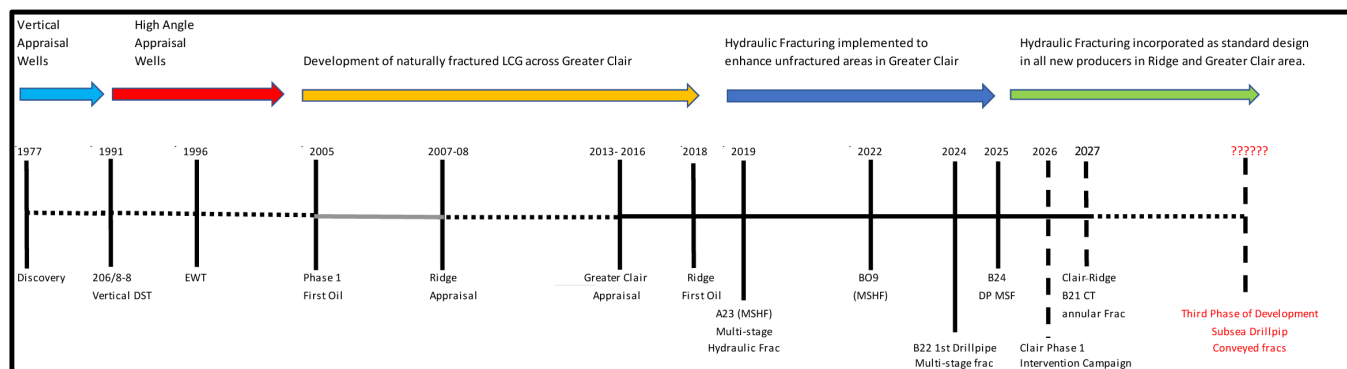


Figure 3—The Greater Clair area development and hydraulic fracturing progression and future potential.

Fig 3 provides the current overview of the Clair Field development progressions, updated from the original produced for SPE 205250 after the first multi-stage frac (MSF) in Clair Phase 1 well A23. Three further MSF wells have now been completed on the Clair Ridge Platform, Clair Ridge wells B09 in 2022, B22 in 2024 and B24 in 2025. The future schedule shows a significant number of planned MSF candidates now hydraulic fracturing is providing a track record of demonstrable uplift in low permeability non-naturally fractured portions of the Clair LCG reservoir. What is also emerging for new drill wells is a standardized completion, based on B22 MSF design utilising drill pipe shifted sleeves which has now also been executed in B24. For existing wells installed during previous drilling campaigns, a variety of MSF techniques will continue to be applied on a case-by-case basis, dependent on the liner design, to optimize the executions.

MSF Installed and Planned Installations:

- Clair A23 and Clair Ridge B09 – MSF installed via CT shifted sleeves (initial rates 6,500 and 4,500bopd respectively).
- Clair Ridge B22 – MSF via drill pipe shifted sleeves (initial rate of 12,500bopd).
- Clair Ridge B24 – MSF via drill pipe shifted sleeves (installed July 2025).
- Clair Ridge B26 – MSF via drill pipe shifted sleeves (to be installed 2026).
- Clair Ridge future producers – all Planned as MSF via drill pipe shifted sleeves.
- Clair Ridge B21 - MSF via CT intervention with CT shifted sleeves (Option for 2027).
- Clair Phase 1 A07 & A15 – CT Intervention based campaign in two wells not originally designed for fracs (Option for 2026).
- Clair Phase 1 – Drilling campaign evaluating 2 new producers with drill pipe shifted sleeve (Option for 2028).
- bp and partners are considering options to unlock future energy potential from the Clair field through a third phase of development. Hydraulic fracturing will be an essential technology, with one potential development option being subsea wells with MSF via drill pipe shifted sleeves.

Significant modeling efforts now underpin the potential value of hydraulic fracturing in the Greater Clair and optimization of the designs, further details can be found in IPTC-22293-MS (Dumitrache, 2022), and SPE-212356-MS (Bird 2023).

The primary focus of this current paper is the design and execution of the completion, as well as fracturing operations in Clair Ridge well B22, which was completed and stimulated in the summer of 2024, and Clair Ridge well B24, which was completed and stimulated in July 2025. A comprehensive operational analysis of the initial MSF via drill pipe shifted sleeves is available in SPE-226610 (van Loon, 2025).

Production Outcomes Through Hydraulic Fracturing

SPE 250250 (Roy, 2022) details how naturally fractured zones and fracture KH dominates wells total deliverability, wells with reduced presence of natural fractures are relatively poor producers. The plan for hydraulic fracturing is therefore to target wells and zones with minimal presence of natural fractures where hydraulic fracturing can increase the production significantly. Hydraulic fracturing provides the ability to establish vertical connectivity in thick reservoir sections between production zones in the horizontal wellbore increasing the total recovery factor in those areas. SPE 215632 (Bird 2023) details how hydraulic fracturing has delivered short term production increases of up to 6.5-fold and sustained 2 or 3-fold increases in the longer term on both Clair Ridge and Clair Phase 1.

The production outcome for Clair Ridge B22 is the subject of further review in the remainder of this paper, in keeping with the outcomes of the previous MSF installations, the post frac initial production for B22 of 12,400bopd has significantly exceeded the predicted upside P10 prediction. Production allocation between hydraulically fracture zones and naturally fractured zones is being evaluated to determine the production uplift from stimulation. At the time of writing, B24 has not been put on production and so no production outcome is available at this stage.

Completion Design

Original Asset Completion Strategy

The asset's completion strategy for producers revolves around three key objectives:

- Construction - sliding sleeve completion with zonal isolation which allows for the combination of naturally fractured zones with hydraulically fractured zones and the ability to change the zonal configuration throughout the life of well.
 - Unit III was historically considered to be sand prone (requiring sand control via standalone screens).
 - Unit V was assessed to be more competent and was historically completed using a (remotely operated) sliding sleeve completion.
 - Both designs utilised swell packers for open hole isolation.
- Stimulation - enabling production from poor quality matrix.
- Life of well management – water and/or gas - breakthrough risks.

Clair Ridge reservoir's naturally fractured structure and low expected fracture gradients make open hole completions ideal, allowing natural fractures to aid in production. This reservoir is divided into two primary units, each functioning as a dual porosity system with favourable permeability, justifying primary development.

The original completion design, consisting of standalone screens for Unit III and pressure pulse sleeves for Unit V, was considered optimal due to anticipated production from the matrix and natural fractures. Initially, proppant stimulation was not deemed necessary to boost production, which saved associated costs. However, consistently targeting natural fractures and high-quality matrix sections has not yielded the expected production, prompting a shift towards proppant stimulation to reduce potential downside outcomes.

← Increasing →		mTVD
Unit III	Unit V	Pay-zone
Lower Strength	Higher Strength	Low Average UCS High
Higher Permeability	Lower Permeability	Low Average Permeability High
No Stimulation Sand control	No sand control Stimulate the well (offline/online)	Historical Completion Design

Figure 4—Sand-face completion strategy.

Reservoir Complexity and its Impact on Completion Design

The standard well placement strategy targets the highest quality rock, requiring multiple assemblies per zone in complex naturally fractured reservoirs, such as swell packers, anchors, production sleeves, stimulation sleeves, and sand control sleeves. The wellbore's nature, including washed-out open holes and minimal shale breaks, along with features like faults and damage zones, necessitates a sophisticated spacing and equipment strategy, with trajectory azimuth and maximum horizontal stress impacting stimulation design.

Equipment Needed to Enable Stimulation

After shifting the strategy to stimulation, mechanical stimulation sleeves were added to Unit V wells, replacing the original remotely operated production sleeves, which necessitated a start-up intervention. Implementing this change was more complex for Unit III wells, which may need sand control. For B22 (placed in Unit III) and B24 (situated in Unit V), an engineered solution was created to facilitate stimulation and remote well start-up in a sand-prone area: remotely operated production sleeves, both with and without sand control technology.

To enable online stimulation activities, relying solely on swell packers to prevent out-of-zone frac initiation was not feasible. The packers would not have swollen to the required differential/anchoring capacity at the start-up stage. Therefore, hydraulically set stimulation packers and anchors were introduced. These packers provided pressure confinement for initiating and generating fracs within the zone of interest. However, the risk of damage to their short elements during screen-out made them sub-optimal for life-of-well isolation. Therefore, the introduction of anchors minimised excessive movement and loads on the completion string. Consequently, swell packers and anchors were used to best enhance life-of-well zonal isolation. Due to the potential for weaker rock in Unit III environments, engineering assessments using simulated rock strength and specific test cells were conducted. This led to the selection of an anchor design with the largest slip contact area for the target reservoir's conditions.

Open Hole Completion Implications on Stimulation Design

Open hole stimulation comes with its own complexities (especially when paired with separate production sleeves): unknown initiation points, relying on open hole isolation to confine the stimulation pressures, and a large annular space where proppant can accumulate during the stimulation operations.

The placement of these assemblies involved reviewing density, porosity, gamma & UCS for sleeve placement (and likely initiation points), and resistivity, saturation, permeability, image & caliper for packer placement. Losses also informed whether the observed features were open or closed. Deep resistivity tools

can highlight potential pockets of hydrocarbons away from the wellbore. Placing sleeves close to the likeliest initiation point in the open hole reduces the annular path between frac and inflow point (to minimise the risk of hole collapse impacting productivity). In a dual porosity system, with weaker strength rocks and pay and non-pay zones, a collaborative and balanced approach is essential for determining the optimal placement of equipment, and the impact on production and stimulation objectives.

New Completion Design - Concept Selection, Equipment Selection and Qualification

Offshore fracturing operations require system flexibility in managing operational risks and mitigations to recover from unplanned events, combined with simplicity and efficiency. Clair does not represent a challenging environment for frac execution per se with a high success rate in achieving the planned proppant placement; stress is mild, there is an absence of strong stress barriers, pressures relatively low and the ability to breakdown formation undisputed. The only challenge to frac execution is the presence of natural fractures which may result in excessive leak off.

The two major operational challenges are posed by the harsh weather environment of the West of Shetland Atlantic and the drilling schedule. When combined, these two aspects may result in a very short weather window available for frac operations. The selection of the completion system for Clair therefore focused on operational efficiency and the ability to recover from screen-outs and/or under-flushed fracs.

A Single Trip MultiStage (STMF) system, [Figure 5](#), was therefore selected for Clair development with fracture execution being preferred during the lower completion phase ("online") to reduce delays in putting wells onto production. For online fracturing the drill pipe frac string, remains in hole for the full duration of the frac operations allowing reversing out the excess of proppant in case of screen out, and enabling under-flushed fracs. STMF applied as online (drill pipe) fracturing is an efficient downhole operation but can be more cumbersome around the transition between stages at surface as a result of the frac iron rig-up and rig-down at each stage. One stage per day can be delivered but the ultimate target is two stages per day as logistics on both the rig and frac vessel are improved.

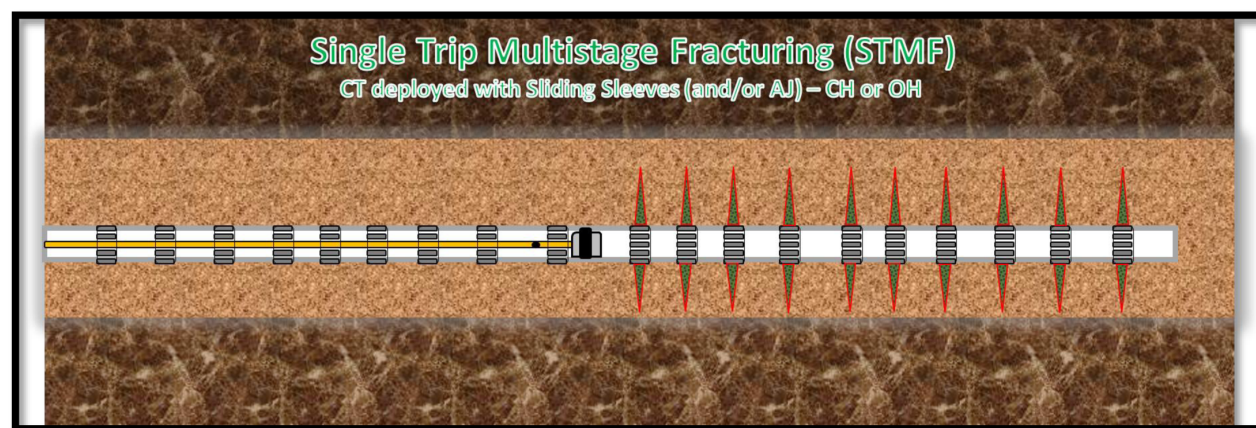


Figure 5—STMF offline annular fracturing simplified schematic ([Casero, 2021](#)).

Pairing Bespoke Production Sleeves with Stimulation Sleeves

Enabling Stimulation in a Historically Sand Control Environment

With B22 targeting the pay zone in Unit III, the historical strategy would have dictated the use of sand screens. However, to achieve well objectives/production enhancement, the completion design requirements were:

- Ability to stimulate.
- Ability to provide sand control down hole if required.
- Ability to remotely open and produce the well, post tree installation.

Bespoke production sleeves with remote technology (similar to the sleeves installed on the asset's injectors), with and without a screen, were designed via a rigorous research and development process. The use of the non-screened production sleeve was implemented to prevent plugging of the short screen shroud on the screened production sleeve during well start-up. Fracability is then achieved via mechanical multistage, single trip fracture system.

However, with the implementation of new equipment and technology, design assurance and system interface testing were of paramount importance. This included computational fluid dynamics (erosion testing, pressure loss analysis, fluid pressures and velocities), inflow performance modelling (production from the stimulated reservoir into the well), material compatibility and design reviews (against API and company policy).

Development of Production Sleeves

With any new equipment to the asset, the design goes through rigorous assurance via: design reviews, internal peer assessments, as well as computational analysis and physical testing.

Significant modelling assessment was completed:

- Erosion of the sand control medium.
- Productivity impacts due to frictional effects through the tool.
- Plugging risk resulting in loss of production and hot spotting risks.
- Fracability and the implications of offset inflow valves from the initiation point of the fracture in an open hole application.

Additional considerations included:

- Operational complexity.
- Modularity.
- Maturity of technology.
- Cost.
- System interface testing.

Use of Computational Fluid Dynamics

Computational fluid dynamics (CFD) was used to assess the erosion and hot spotting, frictional effects from hydrocarbon production through the equipment and plugging likelihood. Expected maximum rates from each producing zone, as well as real worst-case solids loading and sizing, and hydrocarbon viscosity were used to most accurately assure the design.

The sand control filter was not observed as the limiting erosional risk. Rather the inflow ports downstream of the sleeve and micro-annulus were assessed to be at most risk. However, an acceptable 4mm over life of well was observed (for a single sleeve), at expected long term gas-oil ratios (GORs). As expected, by increasing the number of sleeves over the reservoir, production through each sleeve would be split and erosion risks would be reduced. Additionally, the height of the guide wires was increased to further reduce erosion risks.

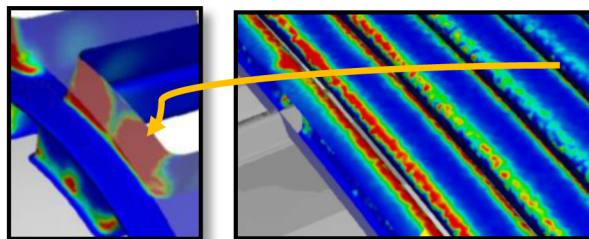


Figure 6—Erosion modelling of screened sleeves indicating highest material reduction at the inflow ports.

Frictional pressure loss was deemed acceptable (6 – 60psi for 5 – 20mbd, respectively). The non-screened sleeves did not present any sizeable frictional pressure loss.

CFD did indicate potential plugging risks during well construction (where a proportion of the flow enters the screen when being run in hole via an induced pressure differential between the open hole and screen x base-pipe annulus).

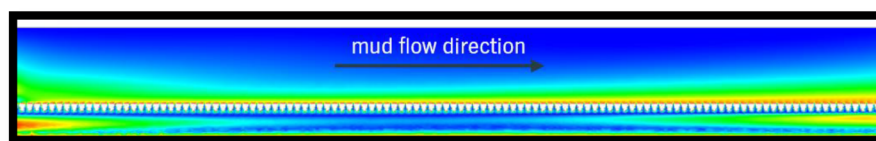
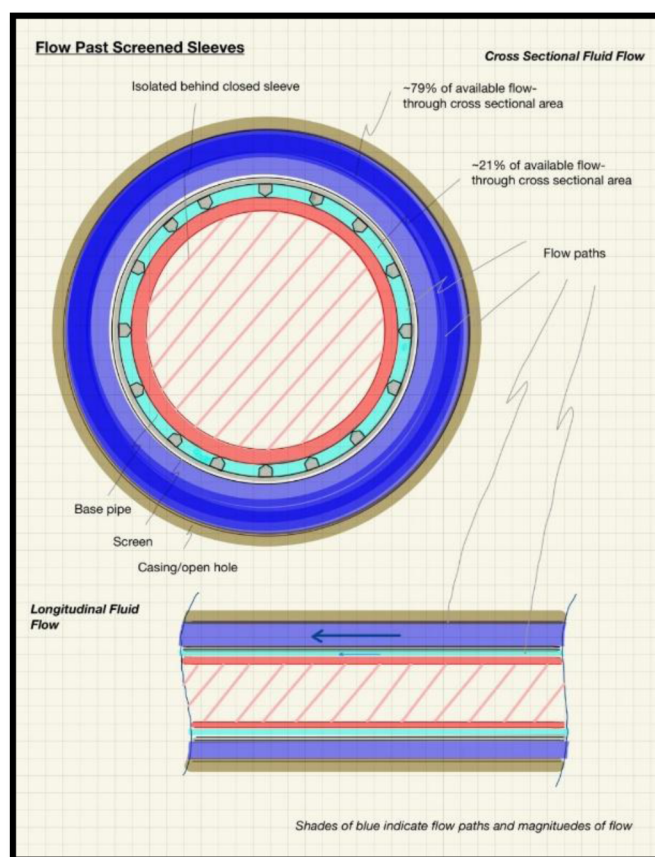


Figure 7—(Above) Sketch of XO-sectional area of screen section of screened sleeve and the ratio of flow areas between the open hole annulus and the screen x base-pipe annulus. (Below) CFD used to assess the generated pressure differential between the two annuli causing flow weaving and potential plugging of the screens during deployment.

Three different mitigations to this risk were introduced: the introduction of solids free oil-based mud (SFOB), stringent production screen testing (PST) and maximising the length of each of the screens. The fluid design was assured by assessing rheology, fluid loss, flow through and compatibility of the fluid systems.

Managing Construction Solids

Over 20+ wells, a significant quantity of solids-laden fluid, as well as lost-circulation material (LCM), has been pumped depending on the presence of natural fractures. If produced through the screened sleeves, this would be expected to flow back and plug off production. Therefore, the non-screened sleeves would be used to start-up the well (allowing it to clean-up), with the option to open the screened sleeves (and close the non-screened sleeves) if significant sanding is observed.

Although the use of a non-screened sleeve does not include sand control, the risk of plugging to the shorter, contiguous screen sections of the sand control sleeves was deemed detrimental to their performance. Therefore, non-screened sleeves were included in the completion design.

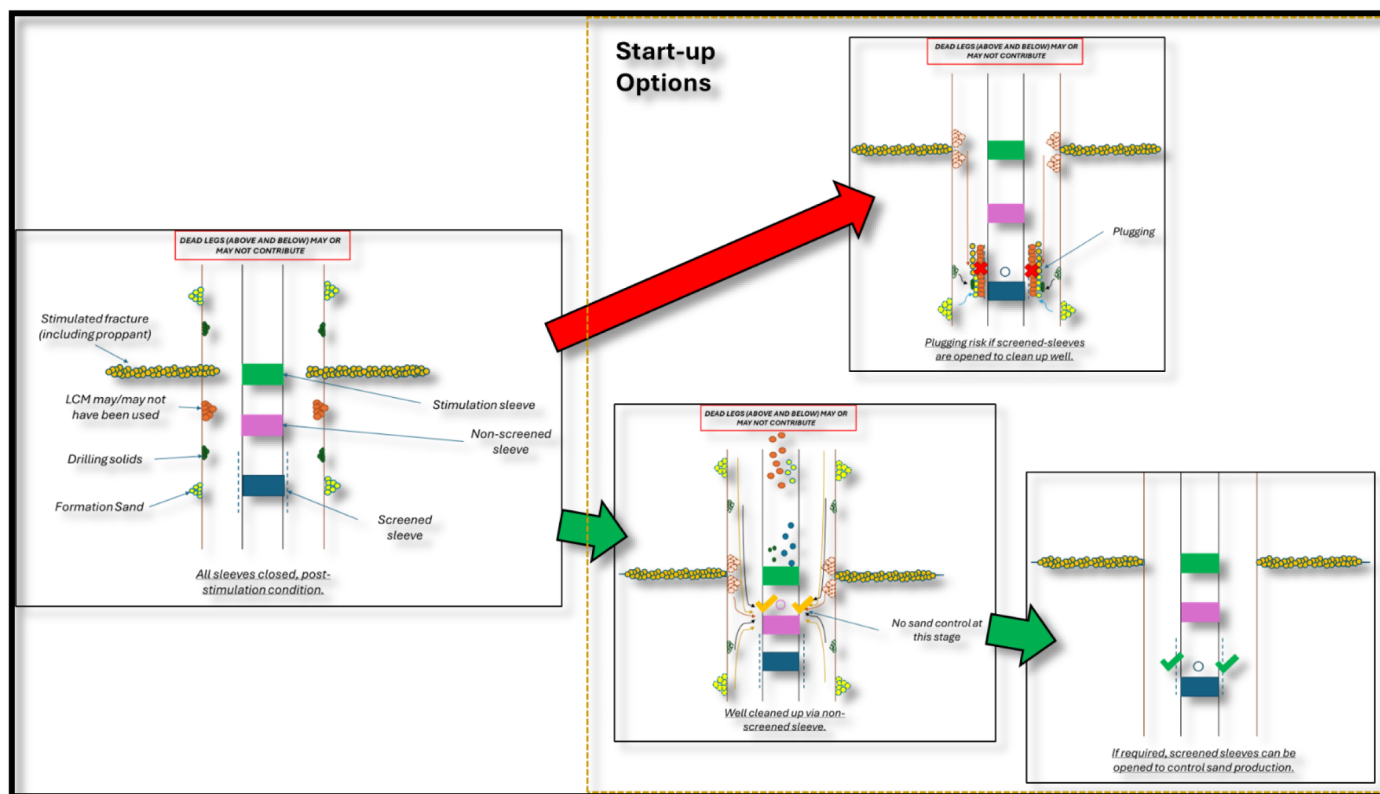


Figure 8—Start-up strategy to minimise plugging risks.

With the design departing from a large quantity of inflow points in a standalone screen scenario (where plugging risks of the screen medium is minimized), the initial sandface approach was to install as many sand control production sleeves as practicable (4x screened sleeves per zone were initially selected).

Sandface Completion Inflow Modelling

The quantity of four sleeves was also selected to minimize the risk of having frac mouths being offset significantly from the production sleeve (in case of hole collapse in the annulus causing significant reduction in production). To visualize this, the following figure was generated, showing how it is most likely that the fracture will generate opposite the stimulation sleeve, and the offset nature of the production sleeves may result in reduced production if hole fill occurs.

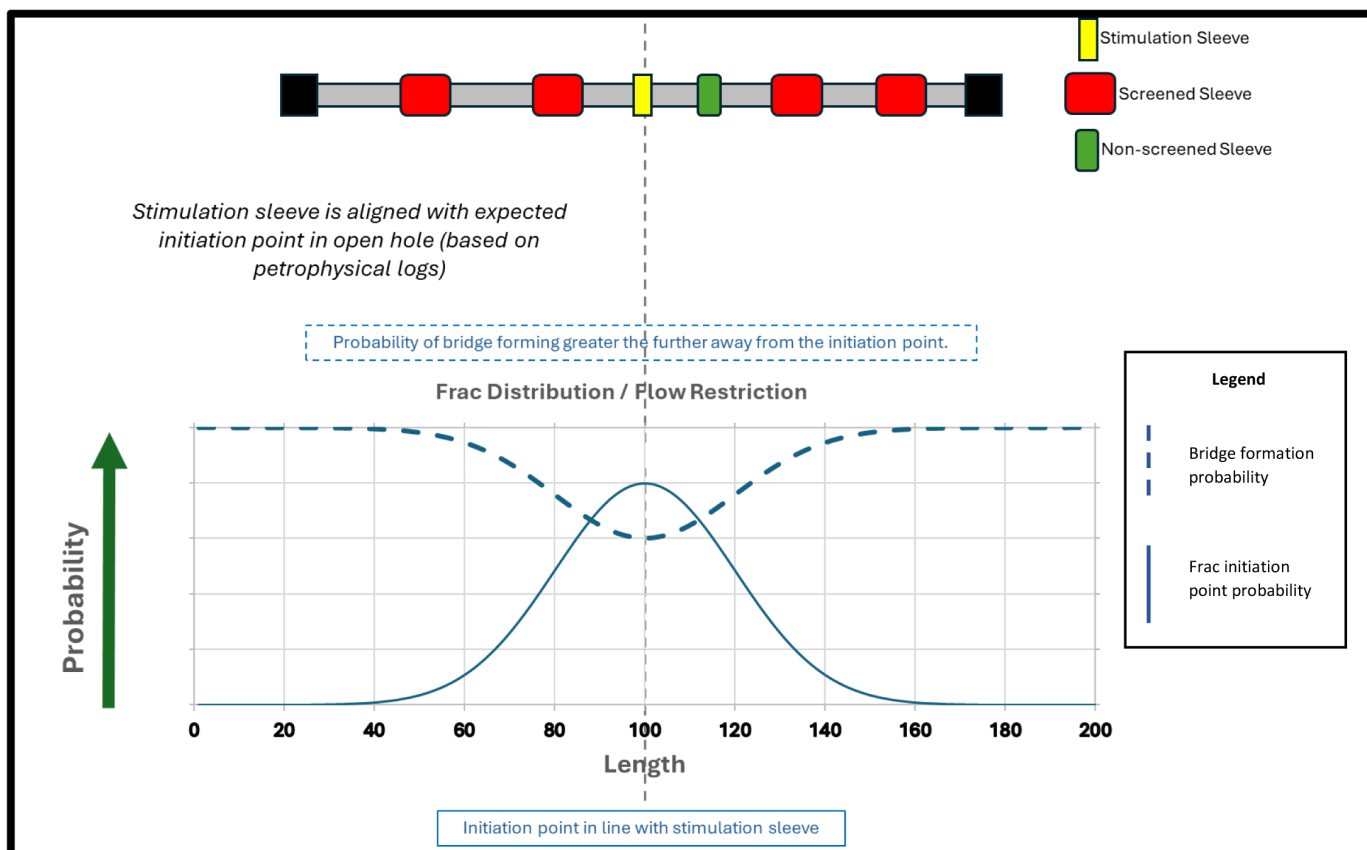


Figure 9—Visual representation of open hole zone, initiation point, hole fill probabilities with initial sleeve configuration.

Wellbore inflow modelling (REVEAL) was used to inform the impact of hole fill (modelled as differing permeabilities to simulate resistances to flow) in aligned and misaligned fracture cases.



Figure 10—REVEAL Modelling highlighted the limited impact on production between the number of sleeves used and misalignment of the fracture, in non-collapsed annulus situations.

Based on offset well analysis of open hole stability of online producers (comparing start-up pressure build-up – PBU – results against later well life conditions), where the risk of hole collapse was assessed

to be minimal (no noticeable reduction in kH), the difference in production between one or two sleeves was minimal.

Initial System Integration Tests of New Production Sleeves (SITs)

For these sleeves, the physical testing related to compatibility between the new production sleeves (both the screened and non-screened sleeves) and the stimulation bottom-hole assembly (BHA). All other equipment had been compatibility checked during previous wells.

The first system interface test completed was to run the stimulation BHA through a production sleeve prototype at ambient conditions. The prototype included the same internal design for both the screened and non-screened production sleeves. This design included an OTIS B shifting profile which seemed to interface with the locating collet in the stimulation BHA. As offset wells had shown it possible to shift a slick bore assembly downhole (with a dual anchor shifting tool), the design was altered to remove any shifting profile. The tool would be planned to be located with casing collar locator (CCL). This was then reassessed with the same testing (again at ambient conditions), which showed that the sleeve did not open when the stimulation BHA passed through.

Management of Risk versus Reward (Assessing the Need for Sand Control and Remote Start-up versus Risk of Inadvertent Sleeve Opening)

B22 Design Addendum

Due to the criticality of the stimulation to the overall well objective, analysis of this risk continued before lower completion deployment. Theoretical and practical testing assessed the likelihood of the risk, but an open production sleeve (above the point of stimulation) would risk the ability to successfully perform the treatment.

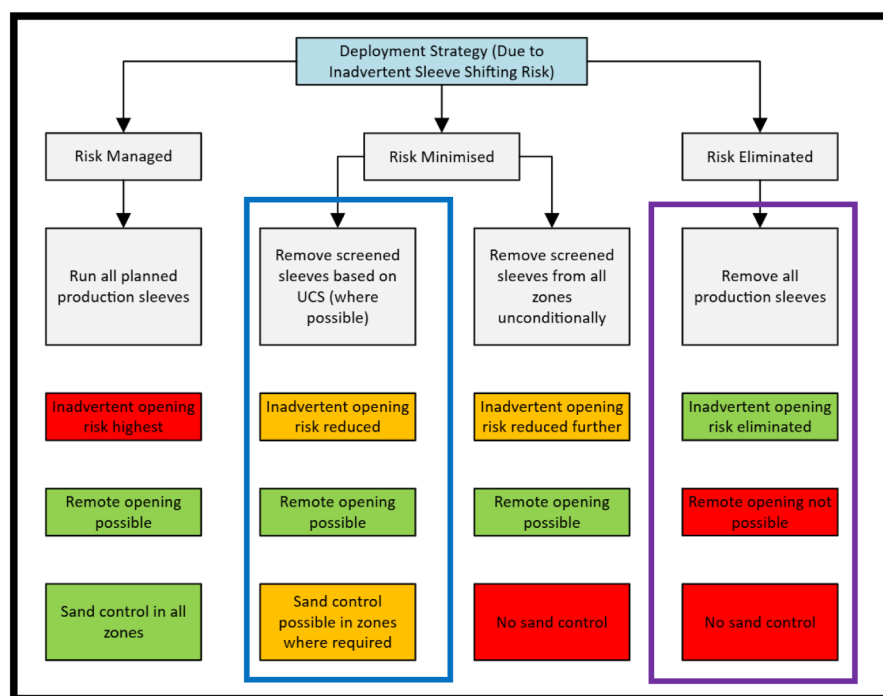


Figure 11—Simplified risk management options. B22 approach highlighted in purple, with the B24 approach highlighted in blue (note that B24 was a Unit V well, where there has been no historical requirement for sand control).

Justification for Sand Control Removal on B22

To further optimise production from Unit III wells, reduce costs, and enable hydraulic stimulation, the need for sand control was assessed again. Comparing assessed rock strength in both Unit III and V wells indicating that a number of Unit V wells had long sections of "weak" rock, that (based on the Unit III sand control criteria) would have dictated the use of screens. However, while these wells do produce more solids than the Unit III wells with sand control, the solids at surface has been more than manageable.

Payzone	Well	Comparative Percentage to Lowest UCS well (Approximate % of weakest Unit III well) <i>Note: Well G is the weakest Unit III well</i>	Comparative Percentage Drawdown to Lowest UCS well (Approximate % of weakest Unit III well)	Comparative Percentage Maximum Production Rate (Approximate % of weakest Unit III well)	Comparative Percentage to Maximum Sand Production To Same Produced Volume (Approximate % of weakest Unit III well)	Comment
Unit V	A	4	32	120	313	
	B	0	60	140	160	
	C	40	128	30	253	
	D	0	124	6	80	
	E	4	96	40	120	
	F	100	124	60	407	Manageable solids production from this Unit V well.
Unit III	G	100	100	100	100	
	H	4	88	140	67	
	I	0	108	40	33	
	J	6	60	70	67	
	K	12	76	60	67	

Figure 12—Comparative analysis of sanding of a number of Unit V wells vs. standard Unit III sand-control wells.

The drilling data acquisition plan for B22 allowed the team to acquire the sonic UCS of the formation. Using this data, historical logs and the regionally agreed sanding risk cut off, there was deemed to be no requirement for sand control. To minimise the risk of inadvertent sleeve opening (by minimising the overall number of failure points), the screened sleeves were removed.

Final System Integration Tests

To better understand shifting forces for the production sleeves downhole (pressure and temperature), a bespoke testing setup was engineered to assess actual shift forces (of a prototype sleeve) that could then be used to inform installed sleeve performance. The conditions were expected to increase seal friction and therefore minimize the inadvertent opening risk.

The results, while in line with what was expected, were nuanced. Initially, when testing at ambient temperature, as the pressure inside the test rig increased, the average seal friction/snap force at each pressure step also increased. However, within these pressure steps there was some variation of results. Tests were continued up to the maximum bottom-hole pressure and then repeated at the maximum bottom-hole temperature. Increasing the temperature did not indicate as clear an increase in stiction (seal friction).

Final Sleeve (Completion) Strategy for B22 and Continuous Improvement for B24

While confidence was gained that down-hole pressure and temperature were increasing the magnitude of force required to shift the sleeves (and therefore there was a lower likelihood of inadvertently functioning), the severity of the risk was still deemed high. Later changes to the test procedure and rig up, resulted in more repeatable results, and indicated significant increases in shifting forces at down-hole conditions. For B22, to protect the ultimate completion objective, a risk elimination approach was taken, and as a result, the production sleeves were excluded from the final installation, relying on the stimulation sleeves for production. Once MSF was proved successful on B22, B24 then introduced these production sleeves (following final assurance of the test results) to eliminate start-up interventions.

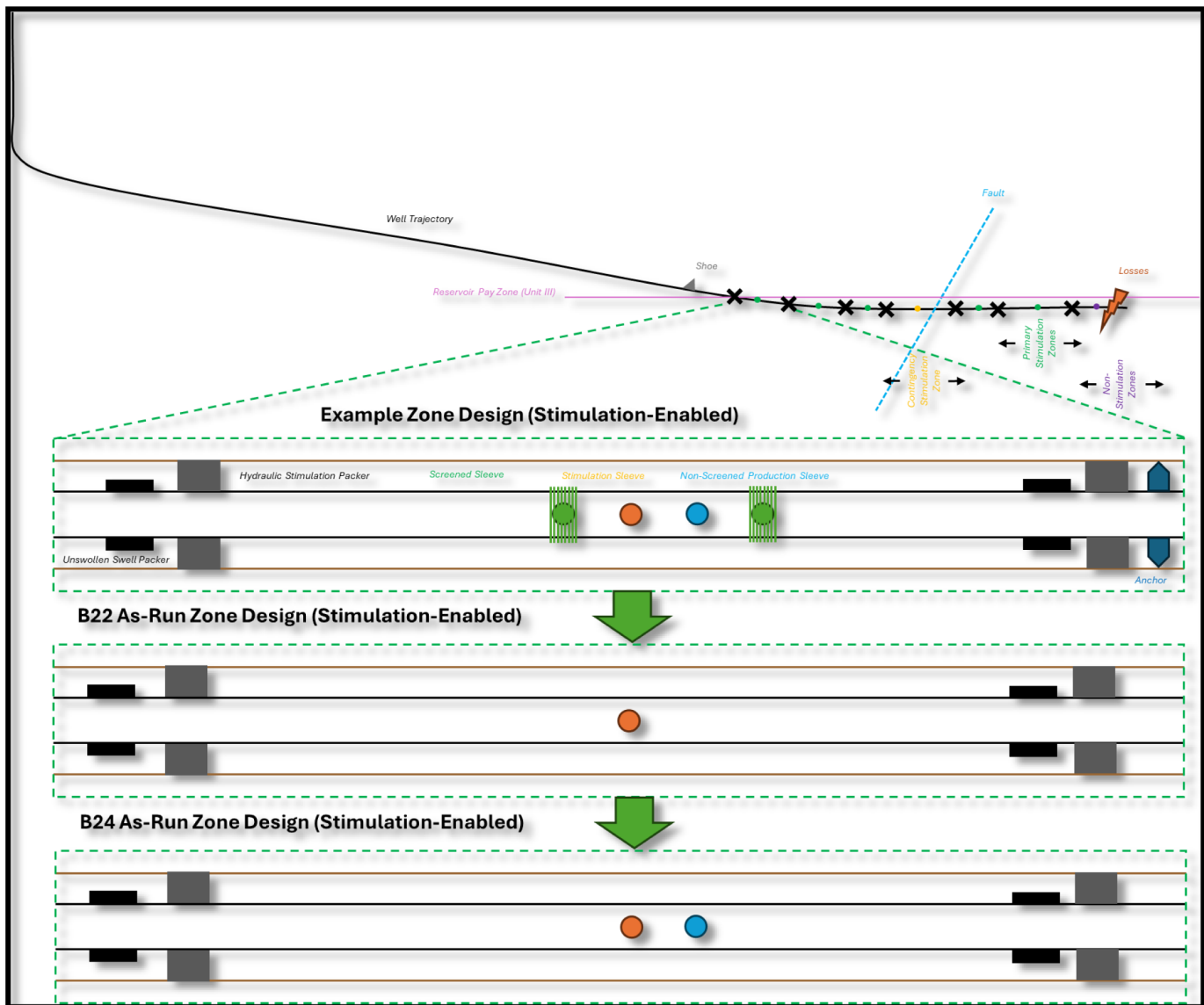


Figure 13—Sandface completion design evolution (design versus B22 versus B24).

The final sleeve strategy for the initial two MSF via drill pipe were as follows:

- B22 was installed with 7 production zones, 5 stimulated.
- B24 was installed with 8 production zones, 6 stimulated.

Online, Multi-Stage, Single-Trip, Hydraulic Fracturing

Platform Rig Up

SPE-226610 (van Loon, 2025) details the rig up for the platform. For online stimulation the line up from stimulation vessel to the frac hangers and up to the rig floor was a bespoke arrangement which also facilitated reversing out the well contents post stimulation to the rig systems. The rig floor manifold connected up to hoses which could move dynamically between the different stimulation positions at each stage (opening the sleeve, treatment, closing the sleeve, reversing out, and then pulling out of hole to the next position). Downhole pressures were monitored via a "dead leg" down the annulus (brine column). Citric acid injection was included in the return system to assist with breaking gel returns.

Hydraulic Fracture Treatments Design

Fracture stimulation strategies in the Clair Field are guided by optimization studies aimed at enhancing fracture conductivity (Bird et al., 2023, Dumitrache et al., 2022). To achieve this, larger proppant, specifically 16/20 mesh, is introduced during the final stages of the treatment to improve the conductivity of the proppant pack near the wellbore, where flow efficiency is most critical. The design also incorporates tip screen-out (TSO) techniques and under-displacement of the fracture at the wellbore to strengthen the connection between the fracture network and the wellbore.

The West of Shetlands (WoS) area lies on the Atlantic Margin and therefore suffers from strong currents, high winds and significant wave heights which dictates April to October is the only suitable period for stimulation operations. The drilling schedule is therefore adjusted to ensure stimulation execution occurs within this period.

Conventional data gathering for stimulation is a luxury in the WoS environment. Diagnostic pumping and decline tests, step rate tests "data" fracs and more, to better calibrate the stimulation models prior to main frac execution were therefore minimized to maximise the operational weather windows. These diagnostic tools are used to gather data, inform geomechanics and better understand the in-situ conditions prior to fracturing were limited to the first stage in every well plus any stages where high fluid loss could be anticipated due to the presence of a fault or natural fracture network. For B22 and B24 the majority of the objectives, usually achieved during conventional data fracs, were therefore delivered via the pad stage, albeit resulting in less accurate leak off data. In addition, there was no calibration of the stimulation model prior to most stimulation stages. With this hybridized fracturing treatment, continuous improvement of the fracture execution timings was observed:

- 13 hours between starting injection for one stage to the next stage where diagnostics were optimised.
- 27 hours between starting injection for one stage to the next stage where a conventional diagnostic program was implemented.

Hydraulic Fracture Treatment Summary

The generalized outcomes of each of the stages for B22 are estimated as per the following table. Reservoir zones #1 (highly naturally fractured zone) and #4 (contingency zone with cemented natural fracture) were not stimulated due to the geological prognosis when drilled.

Table 1—B22 Overall Stimulation Outcome.

Frac Stage	Reservoir Zone	Proppant (klbs)	Estimated Fracture Half-Length, x_f (m)	Fracture Orientation
	#1 (Toe)	N/A	N/A	N/A
#1	#2	400	120	Oblique
#2	#3	500	100	Longitudinal
	#4	N/A	N/A	N/A
#3	#5	500	120	Longitudinal
#4	#6	500	110	Longitudinal
#5	#7	550	130	Longitudinal

The generalized outcomes of each of the stages for B24 are estimated as per the following table. Reservoir Zones #3 (highly naturally fractured zone) and #8 (proximity to gas cap) were not stimulated due to the geological prognosis when drilled.

Table 2—B24 Overall Stimulation Outcome.

Frac Stage	Reservoir Zone	Proppant (klbs)	Estimated Fracture Half-Length, x_f (m)	Fracture Orientation
#1	#1 (Toe)	400	~60	Oblique
#2	#2	400	~60	Oblique
	#3	N/A	N/A	N/A
#3	#4	450	~60	Transverse
#4	#5	450	~60	Oblique
#5	#6	350	~60	Longitudinal
#6	#7	300	~60	Longitudinal
	#8	N/A	N/A	N/A

Post-Stage Clean-Out

After each treatment, the planned ~5bbl of under-flushed 8ppa slurry was reversed out to the rig (down the annulus and up the frac string). Boundary layer modifying friction reducer (polyacrylamide type fluid) was used to manage circulation pressures due to the length of the well. This was added to the brine displaced in and out of the well. On B24, metal-to-metal friction reduction was introduced due to high friction observed at the end of the B22 stimulation operations. Detailed reasoning for this change can be found in SPE-226610 (van Loon, 2025).

The time to start reversing out was minimized to ensure no proppant plug formed, which may cause difficulties reversing out to surface. For both B22 and B24, several assumptions were made to constrain the operational steps for reversing out:

- Sleeves were considered closed prior to reverse out (based on track record) but not proven to be sealing based on pressure testing.
 - To improve the sleeve closure confirmation process for B24, a short, low-pressure confidence test (above initial shut-in pressure) was introduced.
- Downhole pressure was attempted to be kept below fracture closure pressure (FCP) to reduce the likelihood of over-displacing the proppant further into the fracture, in case the sleeve was not closed/sealing. This was achieved by starting to reverse at a lower rate.
 - Analyzing this plan for B22 highlighted that if a sleeve was not closed, this would result in flowback of the proppant back into the well. This was sub-optimal and so the introduction of the confidence test provided better confirmation.
 - With the improved sleeve closure confirmation process for B24, the FCP limit used on B22, was not applicable for B24.
- Potential fluid ingress from the reservoir during reverse out operations was considered a "ballooning" event, as the injected fluids were supercharged into the formation and so would be the expected fluids entering back into the wellbore (not reservoir fluids).

The well was monitored using the rig active system (accounting in variation due to line fill and compression), flow rate in via mud pumps and flow rate out via SLB flow meter, as well as cumulative flow in from the mud pumps and cumulative flow out through the SLB flow meter.

This was not infallible. For B22, when reverse out was conducted with an open sleeve, after the usual 5-10bbl of "apparent" losses, gains were quickly noticeable on rig PVT. However, the well was not shut-in quickly enough, resulting in significant disturbance of the fracture (50bbl of proppant flowed back). This was a key reason to update the reverse out procedure for B24, introducing a confidence test following

attempted sleeve closure. In the event the sleeve was not confirmed closed, the stimulation BHA packer was unset and then reset above the open sleeve, isolating it (as per planned contingency).

The presence of XL Gel across the 3.5" pipe led to pressure "spikes" when ramping up the mud pump to reverse out, due to extra friction. Once the gel started to move, the pressure decreased. This was only visible when XL gel was across the 3.5" string, not once it entered the 5-7/8" DP. The presence of a XL gel "slug" sitting in the 3.5" string led to differences in pressure response between the annulus and drill pipe sides, the more gel in the 3.5" string, the slower the response.

Across both wells, the quantity of under-flushed proppant returned to surface was very variable (for B22 there was between 0.5 to 5mT (wet proppant) returned, compared to a planned weight of ~0.6mT). The injection of 20% citric acid (added at 10gpt), in addition to the line geometry, was efficient to break the gels on B22. The acid injection strategy was established from lab testing to achieve a quick break out of the gel. The reverse line included one strainer downstream of citric acid injection to help with fluid mixing and shear. However, the strainer was bypassed during several reverse-out and did not impact operations.

During reverse out of the last stage, the under-flushed slurry reached surface significantly earlier than anticipated (for B22, equating to ~95bbl, compared with a theoretical volume of ~260bbl). No clear reason has been identified for this but may be in part due to stringing out of the slurry with the low viscosity displacement fluid. This was also observed on B24.

On B22, during the final clean up (performed close to TD, below the bottom stimulation sleeve), there is evidence that the slurry needed a minimum of ~8bpm to be carried to surface throughout all the changes in IDs of the stimulation string. B22 highlighted that the majority of the proppant was removed from the tubing before during the reverse out circulations after every stage. Therefore, B24 removed this operation (saving an entire trip from the operational timings).

Overall Operational Performance

Although this was the first instance of online stimulation performed for the asset and organisation (and a learning curve was expected), the performance improvements implemented to simplify the diagnostic stages of the treatment - as well operational experience gained after each stage - resulted in <24 hours per stage.

Zone	500mz Entry	500mz Exit	Duration
#2	17th Oct 11:25	18th Oct 09:35	~22hrs
#3	3rd Nov 15:15	7th Nov 10:45	~3days 20hrs
#5			
#6			
#7			

Figure 14—B22 Fracturing Execution and Summary.

Reservoir Zone	500mz Entry	500mz Exit	Duration
#1	8 th Jul 14:57	15th Jul 03:27	~6.5days 12hrs
#2			
#4			
#5			
#6			
#7			

Figure 15—B24 Fracturing Execution and Summary.

B22 Flowback, Tracers, Downhole Choke and Production Results

Start-up Strategy

Due to the low reservoir pressure (generating the proppant UCS) and temperature (setting the resin-coated proppant), a careful start-up strategy was utilized to minimize risk of proppant production.

Flowback operations were broken down into phases to manage potential emulsions between any returning OBM and the water-based completion/stimulation fluids.

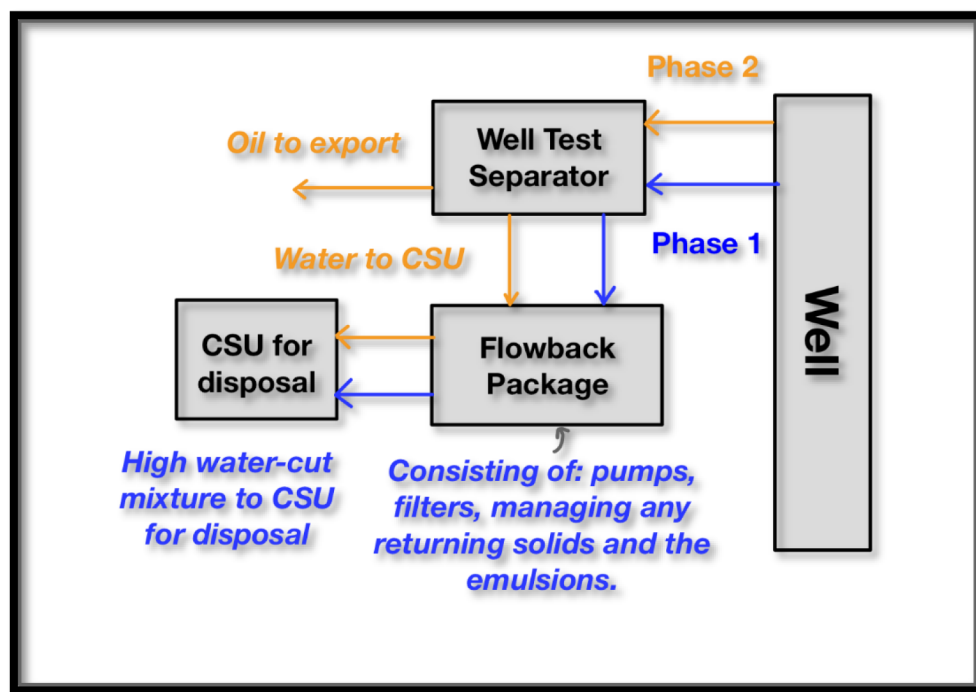


Figure 16—Flowback routing. CSU is the Cuttings Slurification Unit, which prepares fluid for injection into a CRI well (Cuttings Re-injection).

First the annulus was unloaded to gas (for gas lift), after which the well was kick off via the test separator:

1. Phase 1 (all fluids directed to test separator via flowback package):
 - a. The well was brought online to a "low" drawdown and allowed to stabilise. At this point rates, pressures and production index (PI) were assessed. The intent was to step up drawdown (~ 250 psi/day) in stages.
 - i. However, in practice this was not always possible.
 - b. Continued to increase to maximum drawdown over 3 – 4 days.
 - c. Prepared for Phase 2 once water cut dropped to < 500 bwpd (barrels of water per day).
2. Phase 2 (once stimulation fluid flow back dropped below 500bwpd):
 - a. Lined up oil leg to export.
 - b. Continue to assess well productivity through the test separator, whilst batch injecting water into CRI.

Initial Tracer Results

On B22, tracers were only pumped into the stimulated zones (2x zones did not have tracers pumped into them). All zones were open for well start up and so understanding absolute contribution from each zone across the entire wellbore was not possible.

On start-up, at low drawdown and rate, stages 2, 4 and 5, contributed more than stages 1 and 3 (evident from both the gas and oil tracers). As drawdown was kept between 0 and 300psi, production is consistent with the initial contribution split.

After a few days of stable production (and a shut-in period), stage 1 appeared to clean up and contribute the most. Both oil and gas tracer concentration from this stage increase significantly, and acoustic sand detection (ASD) indicates higher solid production to surface. This could be due to increased flow velocities, better carrying the proppant. During start-up, once the ASD drops to baseline levels, the rate of change of production of each of the tracers is consistent. Minimal water is seen in the samples at this stage, indicating the water-cut is coming from the non-stimulated zones (most likely from the toe zone).

Overall, some production was recorded from all stimulated zones during well start-up. However, the 3rd stage has very little contribution throughout, and the 1st stage (at the toe) contributes the most (with its contribution increasing over time).

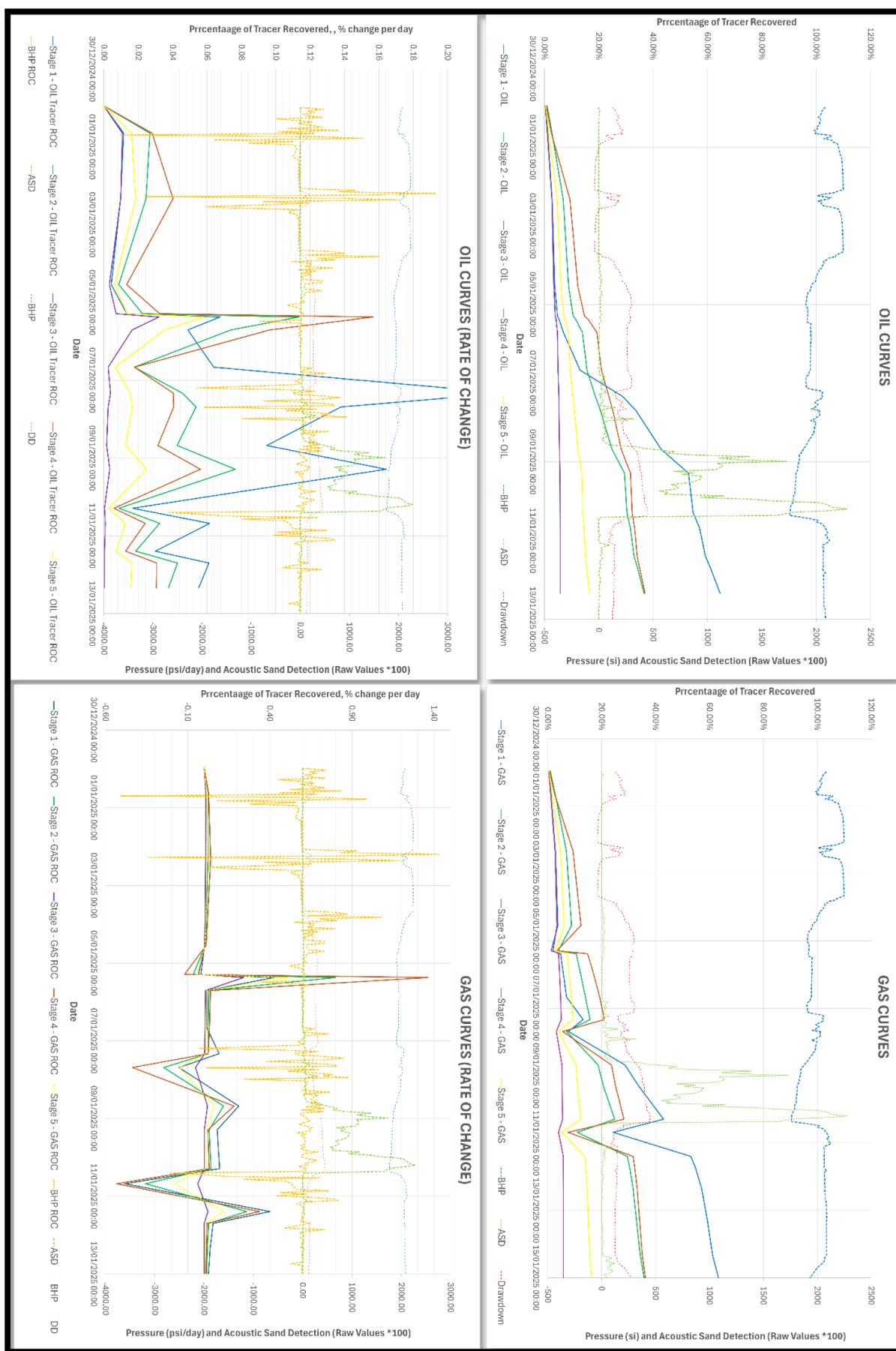


Figure 17—Tracers Oil and Gas curves relative contributions.

Proppant Returns

After well start-up, a total of 11.2t of wet proppant was recovered in the platform test separator, estimated as ~9t of dry proppant (~20% of water in wet proppant). For comparison, this would correspond to an open hole annular volume of ~560m, or ~112m/zone (for 5 zones) of 8ppa slurry. Given the concentration of proppant in 3rd stage influx, there's the possibility that 3rd stage would have contributed to a greater proportion of the proppant recovered at surface. However, given the suggested low rate from that zone based on chemical tracers, there is the possibility that 3rd frac is not contributing and that most of the proppant is coming from the other 4 zones. In this scenario, this would correspond to ~140m/zone (for 4 zones) of 8ppa slurry. This would imply that a significant length of the open hole annular volume was filled with 8ppa in each frac zone, or that (significant) de-hydration occurs in the annulus between the sleeve and the frac sleeve.

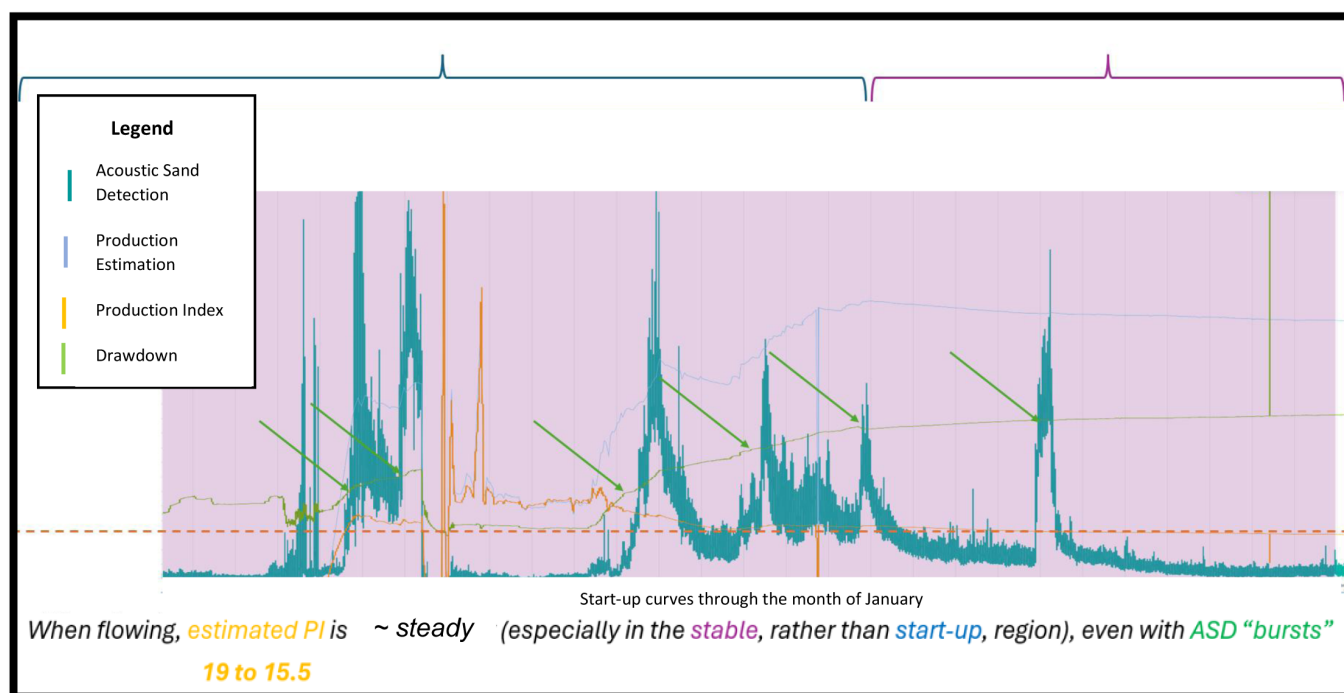


Figure 18—PI evolution with time.

Continuous Improvement for B24

As per the lessons learned flagged in SPE-226610 (van Loon, 2025), the following was implemented in B24:

- Use of strip tracers: the complexity of the reservoir (dual porosity system) means not all zones are stimulation candidates. On B22, the production from these zones (and therefore the contribution of the stimulated zones) was challenging to estimate. To overcome this, B24 strip and inflow tracers to determine contribution of all zones (stimulated or not) to overall well production.
- Flowback package optimisation: a significant rig up is required for a robust flowback package. During B22, this footprint clashed with the online stimulation rig up, and caused logistical/rig-up complexity. B22 confirmed that where significant oil-based mud (OBM) has been lost, emulsions form between the water-based stimulation fluids and the oil-based reservoir drill in fluids (RDIF). These emulsions caused filter blockages that prevented continuous operations during the initial flowback phases. For B24, WBM drilling fluid was introduced and the risk of emulsions has been perceived to be greatly reduced. The flowback package was therefore eliminated to remove significant cost and delay to production start-up, B24 has still to be flowed and cleaned up therefore the outcome has still to be determined.

The Future for Hydraulic Fracturing in the Greater Clair Area

With two Clair Ridge platform wells completed comprising 11 successful stimulation stages Greater Clair has transformed its fracturing completions strategy with a standardized, simplified drillpipe fracturing system. This evolution into efficient multistage fracturing operations has delivered the desired reduction in operational complexity, mitigated the impact on asset schedules and reduced shut-in production. Online stimulation has proven to be less disruptive to the wider asset schedule, require fewer personnel onboard (POB), and present a lower SIMOPs burden compared to offline methods. The future of fracturing in the Greater Clair Area will focus heavily on completing the strategic transition from offline to online stimulation campaigns to stimulate all oil producers during the completion phase, implementing the online system in subsea applications could support unlocking future phases of development.

Optimisation of lower completion and stimulation operations is well advanced, and further progress will continue to improve efficiency by using different combinations of sleeves with various functions. However, the supporting schedule and ancillary operations now offer some of the most significant optimization for stimulation operations and production optimisation.

Well Scheduling

Rig schedules are being optimized to prioritize stimulations during the predictable April to September weather window with the ultimate goal being to deliver batch lower completions and stimulations for two wells in each summer stimulation window. Strategic scheduling particularly through integration and TAR (Turnaround) activities is crucial to deliver overall production optimization for the asset. The focus is on maximising production profiles for the long term.

Catch-up offline stimulation campaigns are also being considered to eliminate backlog and serve as a contingency should online stim strategies encounter execution risks. These campaigns require large amounts of equipment and POB therefore require careful scheduling to slot into the overall platform schedule.

Flowback Management and Infrastructure Enhancements

Currently the well clean-up approach for stimulated wells is limited to all zones being put on production at the same time. The flowback of fluid which may incorporate OBM, LCM, water based frac fluid and reservoir oil, all produced in different proportions at different times from each stimulated/non-stimulated zone, can cause significant disruption to the topsides processing equipment and export systems. Online stimulations inherently introduce unflowed wells to the plant, carrying a mix of water and residual oil-based mud that can risk non-conformance with export specifications.

To mitigate this, dedicated, but temporary, flowback equipment was used to reroute test separator output directly into CRI Well, bypassing the main export path was successfully delivered for B22 allowing the well to be cleaned up without any disruption to the export oil quality. This was however a lengthy process requiring significant equipment and personnel mobilized to the platform. For B24 the reservoir drilling fluid (RDF) has been changed from OBM to WBM with fluid testing indicating this will create less separation issues. The temporary flowback system has therefore been removed and a more conventional start-up will be implemented. The downhole execution of the reservoir drilling and stimulation has been successful with this new fluid system, the production rate and impact on the production facilities has not yet been determined.

Improving Zonal Production Allocation

One of the limitations for Clair is the lack of access on a congested platform to perform PLT activities to help determine the production split between fractured zones, naturally fractured zones and unstimulated zones. A certain amount of production allocation can be determined for the fractured zones in B22 using the tracers deployed in the stimulation fluid. However, without being able to determine the proportion of the total well production which originates from the stimulated zones it is not possible to determine how much production should be allocated to each stimulated zone. To attempt to overcome this limitation, for B24

strip tracer subs have been deployed in all reservoir zones so that the production splits can be determined for all zones. The tracer analysis of the first deployment of the tracer subs across every zone will soon be available once B24 well clean-up has been implemented.

Conclusions and Recommendations

Successful execution of the second STMF well on Clair Ridge has cemented this design as an effective and flexible system to execute hydraulic fracturing in harsh environment such as the North Atlantic West of Shetland. While the system's flexibility has optimised the use the very short windows available late during the year, resulting in placing 5 fracs in 5 days over two weather windows, there is still a significant element of chance around this approach. The B24 well scheduling such that the stimulation execution is in the best weather remains the most robust approach and optimizing the well schedule to deliver summer stimulation campaigns is a key deliverable.

Operational efficiencies will continue to be pursued, however already it is obvious that only marginal gains remain for the online fracturing sequence itself. The largest gains in efficiency remaining will be found in optimizing the reservoir drilling fluid and well clean-up process, alongside further optimizing the well schedule within the overall platform schedule to allow batching of stimulations for two wells to bring similar efficiencies to onshore developments.

While the future of fracturing in the Atlantic Margin and North Sea depends on many factors, STMF is certainly providing a robust technical base for future applications.

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Nomenclature

bbbl	Volume, barrel
bbbl/min	Rate, barrel per minute
BOPD	Oil rate, barrel of oil per day
BWPD	Water rate, barrel of water per day
cP	Fluid viscosity, centiPoise
°	Angle, degree
°F	Temperature, degree Fahrenheit
°C	Temperature, degree Celsius
° API	Oil Density, degree API
ft	Distance, feet
ft-min ^{-0.5}	Leak-off coefficient, feet of fluid penetration per square root minute
gal	Volume, US gallon
gpt	Liquid chemical concentration, gallon per thousand gallons of fluid
in	Diameter, inch
in ²	Flow surface, square inch
kg	Mass, kilogram
km	Distance, kilometre
ksi	Stress, thousand pound per square inch
lb/ft ²	Proppant concentration in the fracture, Pound per square foot
lbm	Pound mass

lbm/kgal	Dry polymer concentration, pound mass per thousand gallons of fluid
lbm/ft	Tubular weight in air, pounds mass per linear foot
m	Distance, meter
md	Permeability, Darcy $\times 10^{-3}$
md-ft	Conductivity, proppant pack permeability in millidarcy x frac width in inch
mesh	Diameter, wires per inch
m ³ /hr	Rate, cubic meter per hour
mm	Distance, meter $\times 10^{-3}$
mMD/RT	Measured depth below rotary table, meter
MMlbm	Mass, million pounds
MMpsi	Pressure, million pounds per square inch
mTVD	Vertical depth below rig Kelly bushing, meter
mTVDSS	Vertical depth below mean sea level, meter
PPA	Proppant stage concentration, pounds mass of proppant in gallon of clean fluid added
ppt	Dry chemical concentration, pound per thousand gal of fluid
psi	Pressure, pound force per square inch
psi BHPg	Pressure, pound force per square inch recorded at the bottom hole pressure gauge
psi/ft	Pressure gradient, pound force per square inch per vertical foot of depth
psi/1000ft	Friction pressure loss gradient, pound force per square inch per linear foot $\times 10^3$
sec-1	Shear rate, reciprocal seconds
sgu	Specific gravity, gram per cubic centimetre
spf	Perforation density, shots per linear foot
t	Mass, kilogram $\times 10^3$
ACA	After closure analysis
BH	Bottomhole
BHA	Bottomhole assembly
BHP	Bottomhole pressure, psi
BHPg	Bottomhole pressure at gauge depth, psi
BHST	Bottomhole static temperature, °F
CCL	Casing-collar locator
CT	Coiled tubing
DP	Dynamic position
EWT	Extended well test
frac	Fracture
F _{cd}	Dimensionless fracture conductivity,
FOI	Fold of increase
FS	Fracture stage
GLV	Gas Lift Valve
geo	Geometric mean
H _f	Fracture height, m
HTHP	High temperature high pressure
ID	Internal diameter, in.
ISIP	Instantaneous shut-in pressure, psi
ISO	International Organization for Standardization
K _H	Horizontal permeability,md
K _V	Vertical permeability,md

KH	formation permeability x sand height, md-ft
LCG	Lower Clair Group
LHS	Left hand side
LT	Low temperature
LW	Light weight
LWP	Light Weight Proppant
NFR	Naturally Fractured Reservoirs
NWB	Near wellbore
NPT	Non-productive time
OIM	Offshore installation manager
OWC	Oil-Water Contact
PAD	Fracture initiation and propagation stage preceding proppant slurry stages
PBR	Polished Bore Receptacle
PBU	Pressure build up
P_c	Closure Pressure, psi
PCA	Post Closure Analysis
PLT	Production logging tool
P_{Net}	Net Pressure, psi
POB	Person on board
POOH	Pull out of hole
RCP	Resin coated proppant
RHS	Right hand side
RIH	Run in Hole
SDT	Stepdown test
S_H	Maximum principal horizontal stress, psi
S_h	Minimum principal horizontal stress, psi
SIT	System Integration Test
SOR	Statement of requirement
STMF	Single Trip Multistage Frac
STOIIP	Stock-tank oil initially in place
SRT	Step rate test
UCS	Unconfined compressive stress, psi
UKCS	United Kingdom continental shelf
WEG	Wireline entry guide
WH	Well head
WHP	Wellhead pressure, psi
X_f	Fracture Half-Length, m

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